



Abhimanyu R B <abhimanyurbsa@gmail.com>

Population pharmacogenomics in Asia-Pacific - referred by Dr. Andrea Gaedigk

8 messages

Abhimanyu R B <abhimanyurbsa@gmail.com>
To: andrew.somogyi@adelaide.edu.au

Sat, May 16, 2026 at 9:28 AM

Hi Professor Somogyi,

Dr. Andrea Gaedigk at PharmVar suggested you might be able to point me toward researchers working on South Asian pharmacogenomics - she mentioned you have a broad view of the Asia-Pacific PGx landscape.

I'm Abhimanyu, building Anukriti - a deterministic pharmacogenomics inference platform focused on underrepresented populations. We implement CPIC guidelines and PharmVar nomenclature across 13 pharmacogenes and 38 gene-drug pairs, with population-aware reasoning and a named-refusal taxonomy: when evidence for a specific ancestry-gene-drug combination is insufficient, we flag it explicitly rather than returning a Eurocentric approximation. South Asian allele characterization is our most critical current gap.

Your 2022 paper on population pharmacogenomics across ten pharmacokinetic genes - and your conference presentation on precision medicine challenges in the Oceania region - suggest you have exactly the cross-regional perspective I'm looking for.

My ask is straightforward: is there anyone you know who is actively working on pharmacogenomic allele characterization for Indian or South Asian populations specifically? We've connected with Dr. Sivasubbu at IGIB on the IndiGen dataset, but the South Asian PGx researcher network seems small and we want to make sure we're not missing key people.

Best,
Abhimanyu R B
Founder, Anukriti
anukritiai.com

Andrew Somogyi <andrew.somogyi@adelaide.edu.au>
To: Abhimanyu R B <abhimanyurbsa@gmail.com>

Sun, May 17, 2026 at 11:45 AM

Dear Abhimanyu R B

Thank you for your enquiry:.

The South Asia countries are Bangladesh, Bhutan, India, the Maldives, Nepal, Pakistan, and Sri Lanka. If that is correct , then I know of only the following who might be in a position to assist
India:

Uppugunduri S Chakradhara Rao

alternate E-mail : rao (at) cansearch.ch , uscrao@jipmer.ac.in

Bhutan: kezangtshe07@gmail.com. (Jigme Dorji Wangchuck National Referral Hospital Changzamtog Thimphu Bhutan 11001)

Kezang is relatively junior as he has applied to Adelaide University to enrol for a PhD in pharmacogenomics.

I do not know of researchers in Bangladesh, the Maldives, Nepal, Pakistan, or Sri Lanka.

Regards
Andrew Somogyi

On 16 May 2026, at 1:28 pm, Abhimanyu R B <abhimanyurbsa@gmail.com> wrote:

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Andrew Somogyi, PhC, DHP, MSc, PhD, FFPMANZCA, FBPhS, FASCEPT
Professor- Discipline of Pharmacology
School of Pharmacy and Biomedical Science
College of Health
Adelaide University
Adelaide 5005
AUSTRALIA
Chair: IUPHAR Pharmacogenetics, Drug Metabolism and Transport Committee
Ph: +61 8 83135572
MOB: 0404498686
Email: Andrew.somogyi@adelaide.edu.au
<http://www.adelaide.edu.au/directory/andrew.somogyi>
CRICOS Provider Number 04249J

Abhimanyu R B <abhimanyurbsa@gmail.com>
To: Andrew Somogyi <andrew.somogyi@adelaide.edu.au>

Sun, May 17, 2026 at 12:17 PM

Hi Prof. Somogyi,

Thank you - I've reached out to both Prof. Rao and Kezang today. I really appreciate you taking the time to map out who's working in this space.

Your observation that the South Asian PGx network is sparse is itself a useful signal for us - it confirms that the data gap we're trying to address is real and largely uncharted.

Since you have such a broad view of the Asia-Pacific PGx landscape, I'd love your perspective on a few things we're working through:

In current clinical trial pipelines that incorporate PGx, what genomic data formats are teams actually submitting: raw VCF, processed diplotype tables, or something else? We want to make sure our output format matches what trial teams can actually consume.

When comparing allele frequencies across populations for a multi-ancestry trial cohort, what are the key constraints or pitfalls to watch out for - particularly when reference panels are Eurocentric?

Is there a minimum viable sample size you'd consider statistically meaningful for estimating allele frequencies in a population like Bhutan or a specific Indian sub-population?

Any perspective you can share would be genuinely useful as we build out the South Asian layer of the platform.

Best,

Abhimanyu R B

Founder, AnukritiAI

anukritiai.com

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Andrew Somogyi <andrew.somogyi@adelaide.edu.au>

Mon, May 18, 2026 at 5:41 AM

To: Abhimanyu R B <abhimanyurbsa@gmail.com>

Dear Abhimanyu R B

To answer your questions- see below.

Regards

Andrew Somogyi

On 17 May 2026, at 4:17 pm, Abhimanyu R B <abhimanyurbsa@gmail.com> wrote:

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Hi Prof. Somogyi,

Thank you - I've reached out to both Prof. Rao and Kezang today. I really appreciate you taking the time to map out who's working in this space.

Your observation that the South Asian PGx network is sparse is itself a useful signal for us - it confirms that the data gap we're trying to address is real and largely uncharted.

Since you have such a broad view of the Asia-Pacific PGx landscape, I'd love your perspective on a few things we're working through:

1. In current clinical trial pipelines that incorporate PGx, what genomic data formats are teams actually submitting: raw VCF, processed diplotype tables, or something else? We want to make sure our output format matches what trial teams can actually consume. IT IS MOSTLY PROCESSED DILOTYPE TABLES PLUS FUNCTION OR ACTIVITY (E.G. POOR METABOLISER)
2. When comparing allele frequencies across populations for a multi-ancestry trial cohort, what are the key constraints or pitfalls to watch out for - particularly when reference panels are Eurocentric?

SOME PANELS ARE NOW GLOBAL PANELS AND INCORPORATE VARIANTS THAT MAY BE RARE IN EUROPEANS BUT MORE COMMON IN SOME JURISDICTIONS (E.G CYP2C19, CYP2C9, VKORCI, HLAS)

3. Is there a minimum viable sample size you'd consider statistically meaningful for estimating allele frequencies in a population like Bhutan or a specific Indian sub-population? NO- AS IT DEPENDS ON THE FREQUENCY OF THE VARIANT BEING TESTED. AS A GUESS SEVERL HUNDRED BUT EVEN IN BHUTAN, THERE ARE 3 DIFFERENT POPULATIONS.

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Abhimanyu R B <abhimanyurbsa@gmail.com>

Mon, May 18, 2026 at 8:13 AM

To: Andrew Somogyi <andrew.somogyi@adelaide.edu.au>

Dear Prof. Somogyi,

Thank you - these are exactly the kind of ground-truth answers that are hard to find in the literature.

A few quick takeaways from our end:

Our current output already matches what you described: we produce metabolizer phenotype calls (PM/IM/NM/RM/UM) with activity scores, so we're aligned with what trial teams actually consume.

The panel selection point is well taken. Our gene coverage already includes CYP2C19, CYP2C9, VKORC1, and HLA-B*57:01. Your note about variants that are rare in Europeans but common in a specific jurisdiction is exactly why we treat population as a first-class input rather than metadata.

The sub-population granularity point resonates strongly. We've already learned not to treat "South Asian" as monolithic; we support GIH, ITU, and PJJ sub-populations, and your Bhutan example is a good reminder that even small countries have internal structure we need to respect.

One more question if I may: In your experience, are CROs running Asia-Pacific trials currently incorporating PGx stratification at the pre-trial stage, or is it still mostly post-hoc analysis?

Warm regards,

Abhimanyu R B

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Andrew Somogyi <andrew.somogyi@adelaide.edu.au>

Mon, May 18, 2026 at 10:02 AM

To: Abhimanyu R B <abhimanyurbsa@gmail.com>

Dear Abhimanyu R B

The HLAs that can cause life threatening drug reactions can be very ancestry specific and with frequencies that vary very widely - so one needs to be quite careful in terms of what HLA test for which drug.

To answer your Q;

are CROs running Asia-Pacific trials currently incorporating PGx stratification at the pre-trial stage, or is it still mostly post-hoc analysis? I DON'T KNOW BUT I SUSPECT IT USED TO BE THE LATTER BUT NOW THAT PGX IS VERY MAIN STREAM ITS MORE AT THE PRETRIAL STAGE. BUT DON'T HOLD ME TO IT.

Regards

Andrew

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Andrew Somogyi <andrew.somogyi@adelaide.edu.au>

Mon, May 18, 2026 at 10:46 AM

To: Abhimanyu R B <abhimanyurbsa@gmail.com>

This might be of interest:

Regards

Andrew

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Huddart_et_al-2019-Clinical_Pharmacology_&_Therapeutics.pdf

498K

Abhimanyu R B <abhimanyurbsa@gmail.com>

Mon, May 18, 2026 at 2:35 PM

To: Andrew Somogyi <andrew.somogyi@adelaide.edu.au>

Dear Prof. Somogyi,

Thank you for sharing this. The Huddart et al. paper is directly relevant to a design decision we've been working through.

We currently use 1000 Genomes superpopulation codes (AFR, AMR, EAS, EUR, SAS) as our population taxonomy. The PharmGKB grouping system is more refined in a few important ways, particularly the SSA/AAC split and the addition of Near Eastern. We'll need to document explicitly where Anukriti aligns with it and where we deliberately differ.

The CYP2C9*8 example in the paper illustrates why this matters. Warfarin dosing algorithms built on *2 and *3 systematically miss the most common reduced-function allele in Sub-Saharan populations. Our named-refusal system is designed to flag exactly this kind of Eurocentric gap rather than silently returning a biased approximation.

I really appreciate you pointing us to this.

Warm regards,
Abhimanyu R B

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